

Rolling Stock Scheduling – Problem Formulation

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1 Problem Overview

Assign N train units to T timetabled trips across S stations over D days. Every trip must be covered, physical continuity must hold, and operational cost minimized. This matches the structure in Guth Jarkovsky et al. (2026) adapted to our 5-station German network.

2 Sets and Parameters

Symbol	Meaning	Code location
U	Set of train units	fleet.csv, fleet_lookup in data_loader.py
T	Set of scheduled trips	trips.csv, trips_lookup in data_loader.py
S	Set of stations	stations.csv
d_{ij}	Distance (km) from station i to j	distance_matrix.csv, dist_lookup
t_{ij}	Travel time (min) from i to j	dist_lookup[(i,j)][0]
cap_u	Passenger capacity of unit u	fleet.csv capacity column
km_max_u	Max km before maintenance	maintenance_rules.csv, maint_lookup
dep_u	Departure minute of trip t	trips.csv departure_min column

3 Decision Variables

Let $x_{u,t} \in \{0, 1\}$ mean unit u is assigned to trip t . In code: chains dict where chains[unit_id] = list of trip_ids.

4 Objective Function

Used in: *src/utls/cost.py*, function *compute_cost()*

Minimize total operational cost:

$$\text{Cost} = w_1 \cdot D_{head} + w_2 \cdot N_{units} + w_3 \cdot V_{maint} + w_4 \cdot P_{uncov} + w_5 \cdot V_{time}$$

Where:

- D_{head} = total empty (deadhead) kilometers traveled
- N_{units} = number of train units activated
- V_{maint} = total km exceeding maintenance threshold
- P_{uncov} = total uncovered passenger demand
- V_{time} = number of timing constraint violations
- Weights: $w_1 = 1, w_2 = 150, w_3 = 5, w_4 = 2, w_5 = 1000$ (DEFAULT_WEIGHTS in cost.py)

5 Constraints

5.1 Trip Coverage

Every trip must have enough capacity:

$$\sum_{u \in U} x_{u,t} \cdot cap_u \geq demand_t \quad \forall t \in T$$

Checked in: compute_cost() via capacity_supplied dict

5.2 Continuity Constraint

Unit u can start trip t_2 only after finishing t_1 and traveling to the new origin:

$$arr_{t_1} + turnaround_{t_1} + t_{dest(t_1), orig(t_2)} \leq dep_{t_2}$$

Checked in: src/utls/feasibility.py, function is_feasible_arc()

5.3 Maintenance Constraint

Total km for unit u must not exceed maintenance threshold:

$$\sum_{t \in chain_u} (d_{head,t} + dist_t) \leq km_{max_u}$$

Tracked in: compute_cost() via km_since_maint accumulator

6 Connection to Paper

This formulation directly mirrors the rolling stock planning problem in Guth Jarkovsky et al. (2026), where the objective is to minimize empty kilometers while satisfying trip coverage, continuity, and maintenance constraints. Their QAOA approach encodes these constraints as a QUBO energy landscape. Our algorithms (Greedy, SA, GA, NSGA-II) solve the same formulation classically, establishing baselines before the QUBO quantum phase.